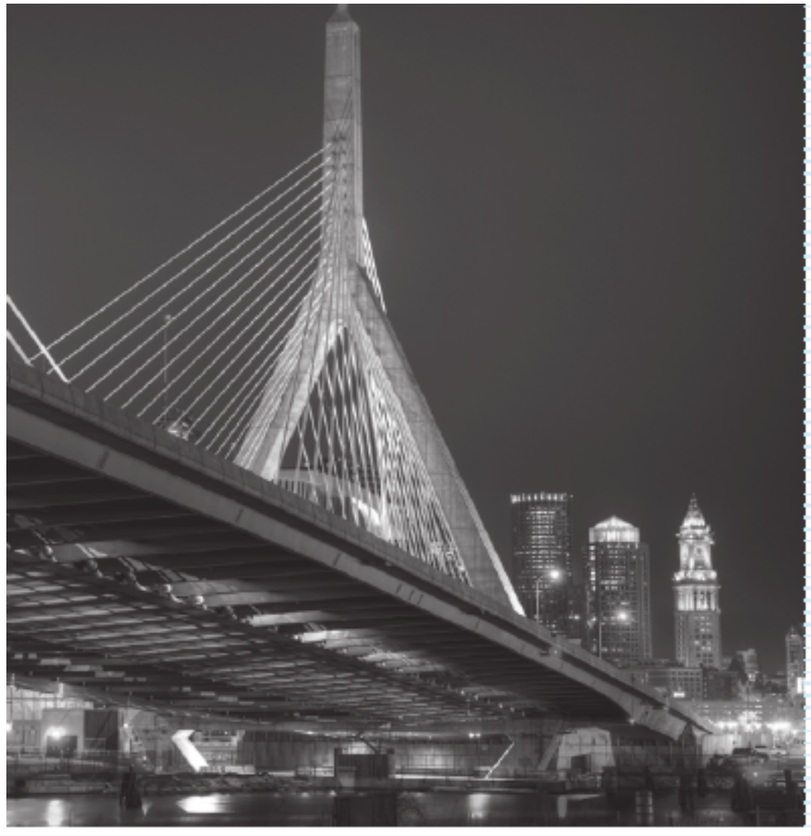


PART A

Ordinary Differential Equations (ODEs)

CHAPTER 1 First-Order ODEs
CHAPTER 2 Second-Order Linear ODEs
~~CHAPTER 3 Higher-Order Linear ODEs~~ x
CHAPTER 4 Systems of ODEs. Phase Plane. Qualitative Methods
CHAPTER 5 Series Solutions of ODEs. Special Functions
CHAPTER 6 Laplace Transforms

one section only

CHAPTER 1 First-Order ODEs

1

1.1 Basic Concepts. Modeling

1.2 Geometric Meaning of $y' = f(x, y)$. Direction Fields, Euler's Method

1.3 Separable ODEs. Modeling

1.4 Exact ODEs. Integrating Factors

1.5 Linear ODEs. Bernoulli Equation. Population Dynamics

~~1.6 Orthogonal Trajectories. *Optional*~~

~~1.7 Existence and Uniqueness of Solutions for Initial Value Problems~~

Whats an Ordinary Differential Equation?

Whats an equation?

LHS = RHS
always we have an
an unknown to eqn

$2x = 5$

$y^2 + 2 = \sin x$

$$\boxed{y^2 + 2\frac{dy}{dx} = \sin x}$$

$\Downarrow$

ODE: equation : solving for something "y"

↳ in ODE we will be solving for a function

↳ in ODE derivative of unknown must be present

Typical ODEs we will be dealing with

find $y(x)$,

$$\rightarrow \quad \frac{dy}{dx} = y^2 + \frac{d^3}{dx^3}(e^x)$$ 1st order ODE

$$\rightarrow \quad 9\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + y = \sin(x)$$ 2nd order ODE

Differential Equations

ODE

e.g. $\frac{d^2y}{dx^2} + y = e^x$

unknown is one variable function

Partial differential eq^n (PDE)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-(x^2+y^2)}$$

unknown is a multivariable function

Why should we care about learing ODE?

because they pop up everywhere

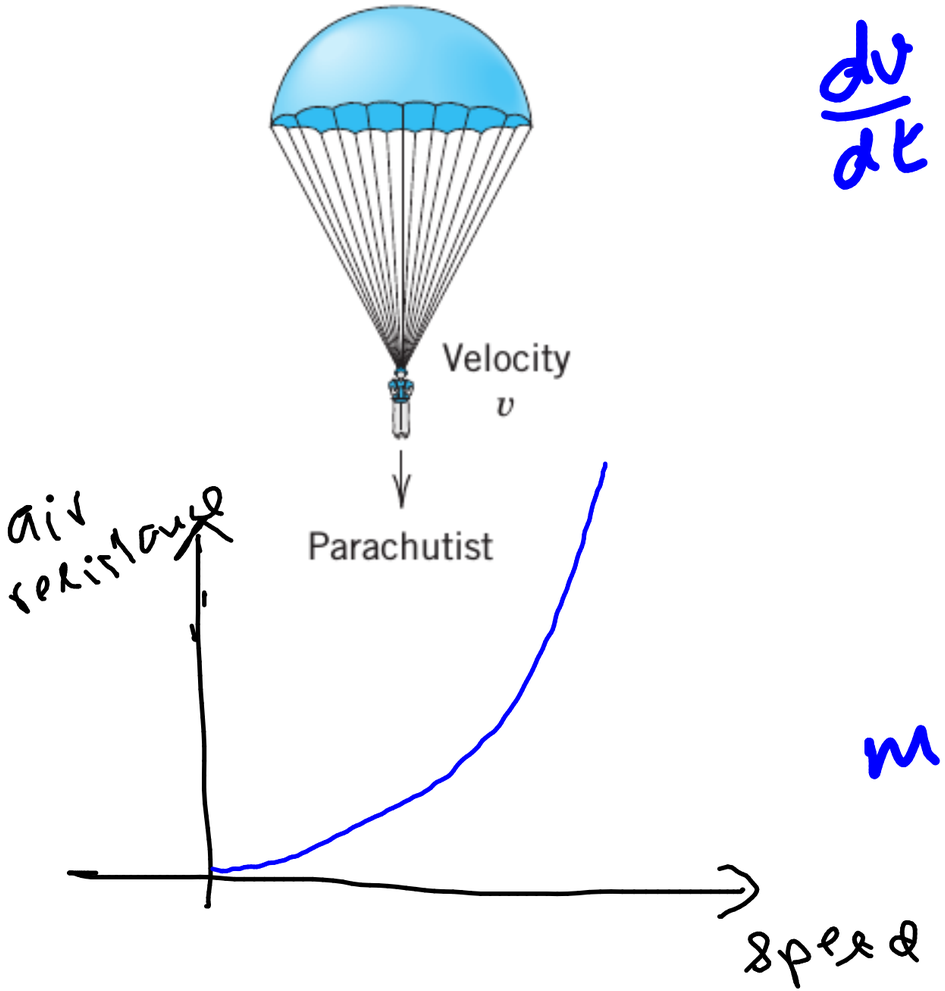
Unknown:

$v(t)$: velocity at time t

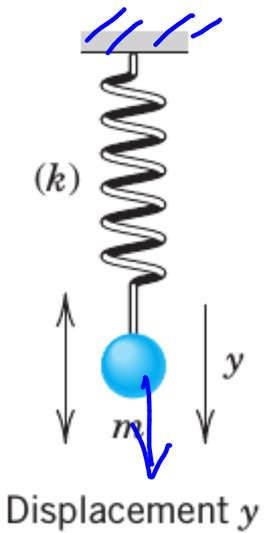
$\frac{dv}{dt} = v'(t)$: acceleration

$$\underbrace{mv' = mg - \underbrace{bv^2}}_{F = ma}$$

mass × acceleration = net force



$y(t)$: position of mass at time t



Vibrating mass on a spring

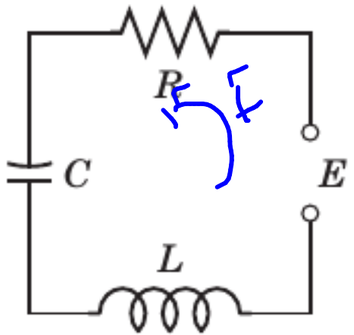
$$\overbrace{my''}^{ma} + ky = 0$$

order 2

$F = ma$

$my'' = -ky$

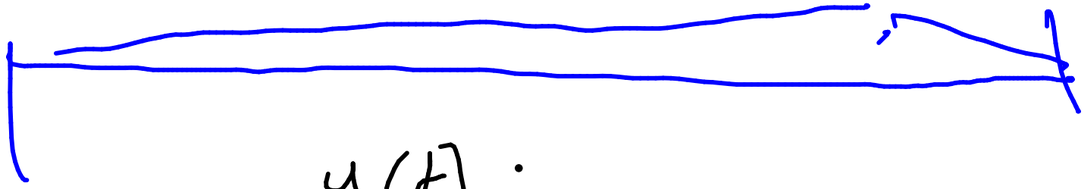
$I(t)$: current in circuit at time t



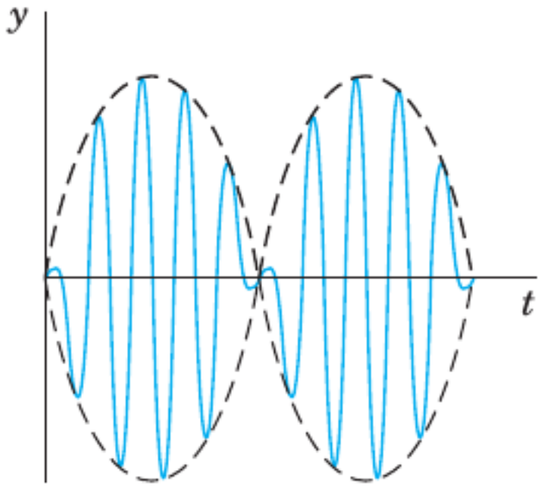
Current I in an RLC circuit

$$LI'' + RI' + \frac{1}{C} I = E'$$

Kirchoff law implies



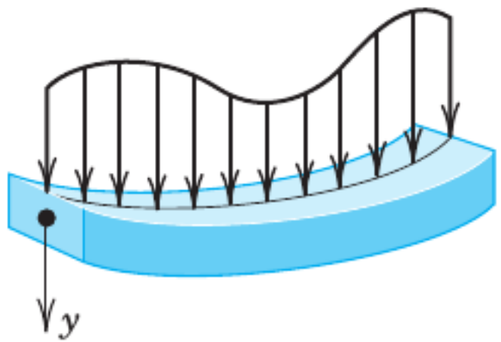
$y(t)$:



Beats of a vibrating system

$$y'' + \omega_0^2 y = \cos \omega t$$

ODE comes into play in study of music

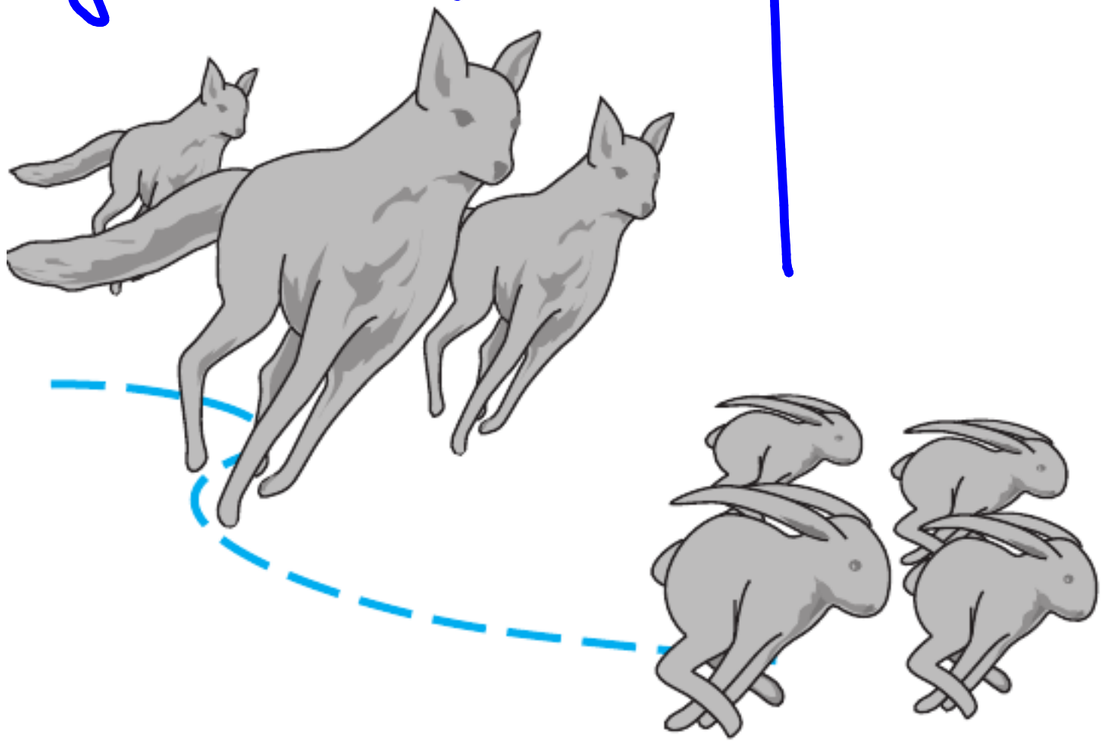


Deformation of a beam

$$EIy^{iv} = f(x)$$

$$\underbrace{EI}_{\text{material elasticity}} y^{(4)} = \left[\begin{array}{c} \text{external} \\ \text{Load} \end{array} \right]$$

system of ODE



Lotka–Volterra
predator–prey model

$$y_1' = ay_1 - by_1y_2$$

$$y_2' = ky_1y_2 - ly_2$$

$y_1(t)$: population of prey

$y_2(t)$: population of predator

In this course we will mostly learn about solving differential equations.

We will not be concerned too much about how these equations came into existence

Easy equations can be solved by

Separation of variables

e.g. find $y(x)$ s.t.

$$\frac{dy}{dx} = \cos(x)$$

$$\left[y(x) = \sin(x) + C \right]$$

Q: Solve for $y(x)$ $\quad xy' = -y$

$$\frac{1}{y}\frac{dy}{dx} = -\frac{1}{x}$$

$$\frac{1}{y}dy = -\frac{1}{x}dx$$

$$\int \frac{1}{y}dy = \int -\frac{1}{x}dx$$

$$\ln y = -\ln x + c$$

$$\ln(yx) = c$$

$$yx = c$$

$$\boxed{y = \frac{c}{x}}$$

check that

$$y = \frac{c}{x}$$

satisfies the given ODE

$$\cancel{x}\left(\frac{-c}{x^{\cancel{2}}}\right) = -\frac{c}{x}$$

EXAMPLE 4 **Initial Value Problem**

Solve the initial value problem

$$\frac{dy}{dx} = 3y, \qquad y(0) = 5.7$$

$$\frac{1}{y}\,dy = 3\,dx$$

$$\ln y = 3x + C$$

$$y = e^{(3x+C)}$$

general soln

$$\boxed{y(x) = C e^{3x}}$$

$$5.7 = C(1)$$

find C using the criteria $y(0) = 5.7$

$$Ce^{3(0)} = 5.7$$

$$C = 5.7$$

Particular solution

$$\left[\, y = 5.7 e^{3x} \,\right]$$

EXAMPLE 4 **Initial Value Problem**

Solve the initial value problem

$$y' = y - y^2 \qquad y(0) = 0.25$$

$$\frac{dy}{dx} = y - y^2$$

$$\frac{1}{y-y^2}\,dy = dx$$

$$\int \frac{dy}{y-y^2} = \int dx$$

$$\ln\left(\frac{y}{1-y}\right) = x + C$$

$$\frac{1}{y-y^2} = \frac{1}{y} + \frac{1}{1-y}$$

$$\int \frac{1}{y-y^2}\,dy = \int \frac{1}{y}\,dy + \int \frac{1}{1-y}\,dy$$

$$= \ln(y) - \ln(1-y)$$

$$= \ln\left(\frac{y}{1-y}\right)$$

$$\frac{y}{1-y} = e^{(x+C)} = Ce^{x}$$

find C

$$\frac{0.25}{.75} = C = \frac{1}{3}$$

Q: find general solution of

$$y' = 1 + y^2$$

$$\frac{1}{1+y^2}\,dy = dx$$

$$\tan^{-1}(y) = x + c$$

$$\boxed{y = \tan(x+c)}$$

Initial Value Problem (IVP). Bell-Shaped Curve

Solve $y' = -2xy$, $y(0) = 1.8$.

$$\frac{dy}{dx} = -2xy$$

$$\frac{1}{y}\,dy = -2x\,dx$$

$$\ln y = -x^2 + c$$

$$y = ce^{-x^2}$$

general sol^n

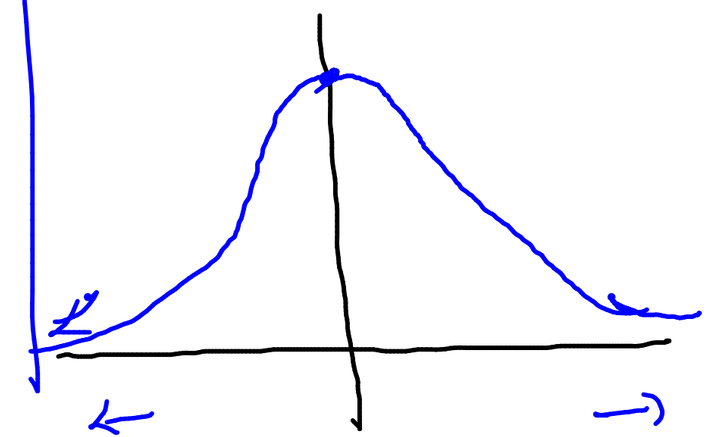
$$y = ce^{-x^2}$$

using $y(0) = 1.8$

$$y(x) = 1.8e^{-x^2}$$

Solve & plot the sol^n:

graph of e^{-x^2}



Last time

→ introduce ODE

differential

ordinary derivative

Equation

→ Started solving ODE

→ 1st method → separation of variables

Reduction to Separable Form

Solve

$$\boxed{2xy\frac{dy}{dx} = y^2 - x^2}$$

?? $\rightarrow$ (only y terms) dy = (only x terms) dx

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

$$\frac{dy}{dx} = \frac{\left(\frac{y}{x}\right) - \left(\frac{x}{y}\right)}{2}$$

change variable

$$v = \frac{y}{x}$$

use $y = vx$ to eliminate y from given ODE

$$\boxed{\begin{aligned} y &= vx \\ \frac{dy}{dx} &= v + x\frac{dv}{dx} \end{aligned}}$$

$$\frac{dy}{dx} = \frac{v - \frac{1}{v}}{2}$$

$$v + x\frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

Solve this ODE for v

$$x\frac{dv}{dx} = -\frac{(1+v^2)}{2v}$$

$$\frac{2v}{1+v^2}\,dv = -\frac{1}{x}\,dx$$

& integrate

$$\ln(1+v^2) = -\ln x + C$$

$$1+v^2 = \frac{C}{x}$$

& finally solve for y

$$1+\frac{y^2}{x^2} = \frac{C}{x}$$

Solution of the ODE is y which is defined implicity by this eq^n

$$\boxed{y^2 = Cx - x^2}$$

EXAMPLE 5 Mixing Problem

Mixing problems occur quite frequently in chemical industry. We explain here how to solve the basic model involving a single tank. The tank in Fig. 11 contains 1000 gal of water in which initially 100 lb of salt is dissolved. Brine runs in at a rate of 10 gal/min, and each gallon contains 5 lb of dissoved salt. The mixture in the tank is kept uniform by stirring. Brine runs out at 10 gal/min. Find the amount of salt in the tank at any time t.

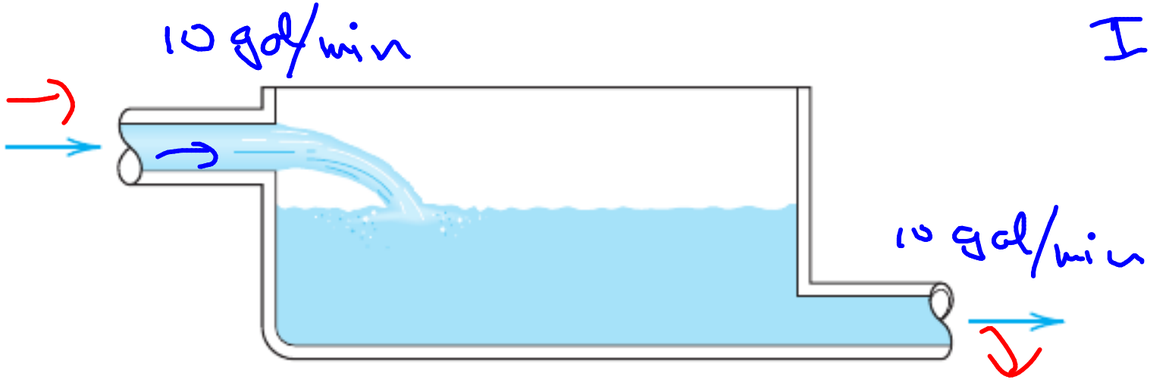


Fig 11

Initial values: 1000 gal water
100 lb salt

Let $y(t)$ = mass of salt present in the tank at time t

$[y(0) = 100]$

(rate of change of salt mass in tank) = (inflow rate of salt) - (outflow rate of salt)

mathematical model

$$\left.\begin{aligned} \frac{dy}{dt} &= 50 - y\,\frac{10}{1000} \\ y(0) &= 100 \end{aligned}\right] \circledast$$

[at time t
$y(t)$: mass of salt in tank
$\frac{y(t)}{1000}$: mass of salt per gallon in tank]

work for y:

solve $\circledast$, plot the solution, analyze the solution

$$\frac{dy}{dt} = \frac{5000 - y}{100}$$

$$\frac{1}{5000 - y} = \frac{1}{100}\,dt$$

$$\rightarrow \quad y(t) = 5000 - 4900\,e^{-\frac{t}{100}}$$

Solution:

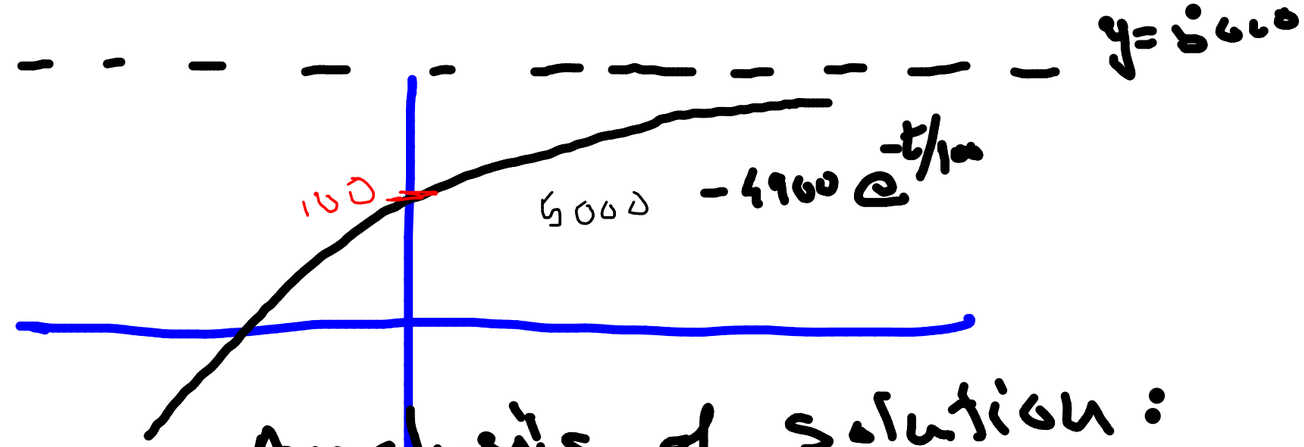
$$y(t) = -5000 \quad 4900\, e^{-\frac{t}{100}}$$

graph]: e^{t}

e^{-t}

$e^{-t/100}$

$-4900\, e^{-t/100}$



Analysis of solution:

Salt steadily increases from 100 lb & stabilizes at 5000 lb

EXAMPLE 6 Heating an Office Building (Newton's Law of Cooling[3])

Suppose that in winter the daytime temperature in a certain office building is maintained at 70°F. The heating is shut off at 10 P.M. and turned on again at 6 A.M. On a certain day the temperature inside the building at 2 A.M. was found to be 65°F. The outside temperature was 50°F at 10 P.M. and had dropped to 40°F by 6 A.M. What was the temperature inside the building when the heat was turned on at 6 A.M.?

Physical information. Experiments show that the time rate of change of the temperature T of a body B (which conducts heat well, for example, as a copper ball does) is proportional to the difference between T and the temperature of the surrounding medium **(Newton's law of cooling).**

$$\boxed{\frac{dT}{dt} = k(T_A - T)}$$

$T(t)$ = temperature at some given point at time t

comments about this equation

- heat exchange between bodies in contact is proportional to difference in temperature

given:

T_A: Temperature of atmosphere

k: heat exchange coefficient

- solve & analyze:

Solve

$$\frac{dT}{dt} = k\left(T_A - T\right)$$

$$\& \quad T(0) = T_0$$

$$\frac{1}{T_A - T} = k\,dt$$

$$\ln|T_A - T| = -kt + c$$

$$T_A - T = c\,e^{-kt}$$

$t = 0$

$$T_A - T_0 = c$$

$$T(t) = T_A - (T_A - T_0)e^{-kt}$$

$$\lim_{t\to\infty} T(t) = T_A \qquad (\because k > 0)$$

Analyze solution

→ $\lim_{t\to\infty} T(t) = ??$

→ why this limit makes sense

body temp will eventually
match surrounding temperature

"Euler method" for solving $\frac{dy}{dx} = \sin(x)$, $y(0) = 1$

[we know $y = -\cos(x) + 2$]

→ decide on which interval we want to solve

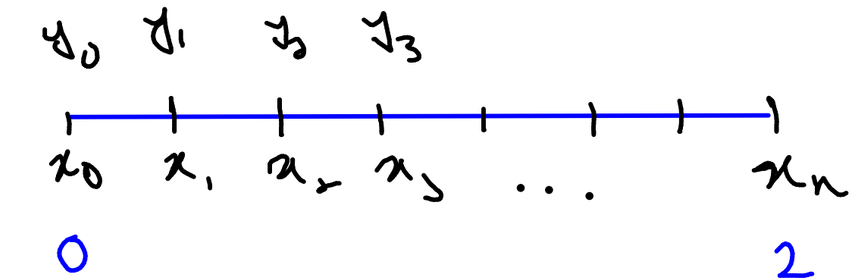
→ find approximate values of y at finitely many discrete points

→ find $y_i \approx y(x_i)$

→ create difference equation for y_i

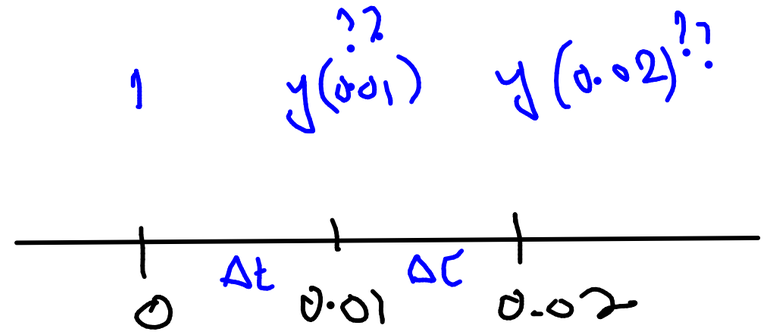
$$\frac{dy}{dx} = \sin(x)$$

$[0, 2]$



$$\frac{dy}{dx} = \sin(x)$$

$$\approx$$



$$\frac{y_1 - y_0}{\Delta t} = \sin(0) \rightarrow y_1 = 1 + (\Delta t)\sin(0)$$

$$\frac{y_2 - y_1}{\Delta t} = \sin(0.01) \rightarrow y_2 = y_1 + \Delta t \sin(0.01)$$

Once again

$$\frac{dy}{dx} = (x^2+y^2) \quad , \quad y(0) = 0.5$$

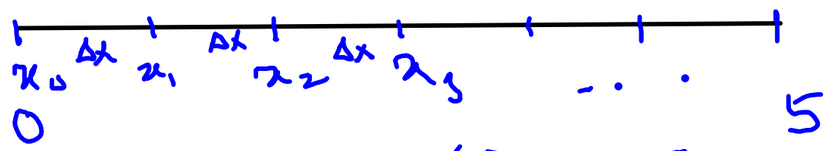
solve using Euler method on $[0, 5]$

$$\frac{y_1 - y_0}{\Delta x} = x_0^2 + y_0^2$$

$$y_1 = y_0 + \Delta x\,(x_0^2 + y_0^2)$$

$$y_2 = y_1 + \Delta x\,(x_1^2 + y_1^2)$$

$y_0 \quad y_1 \quad y_2 \quad y_3$



$$y_0 = y(0) = 0.5$$

$$y_1 \approx y(x_1)$$

$$\ldots \quad y_i = y(x_i)$$

Finding approximate solution

"Euler method"

$$y' = (y - x)^2, \quad y(0) = 0$$

modify the provided matlab code to solve this eqn using Euler method

$$\sinh(x) = \frac{e^{x} - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^{x} + e^{-x}}{2}$$

Exercise:

reduce number of x_i

d. modify the provided matlab script to solve

$$y' = -5x^4y^2, \quad y(0) = 1$$

Sol. $y = 1/(1 + x)^5$